

TUGAS 5

No. _____

Date: _____

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☐ Jawaban

☐ 1. Mencari nilai turunan

☐ (a) $f(x) = \left(\frac{1}{x^2} - \frac{3}{x^4} \right) (x + 5x^3)$

☐ $f'(x) = \left(\frac{d}{dx} \frac{1}{x^2} - \frac{3}{x^4} \right) (x + 5x^3) + \left(\frac{d}{dx} (x + 5x^3) \right) \left(\frac{1}{x^2} - \frac{3}{x^4} \right)$

☐ $= \left(\frac{-2}{x^3} + \frac{12}{x^5} \right) (x + 5x^3) + (1 + 15x^2) \left(\frac{1}{x^2} - \frac{3}{x^4} \right)$

☐ $= \left(\frac{-2}{x^2} + \frac{12}{x^4} - 10 + \frac{60}{x^2} \right) + \left(\frac{1}{x^2} + 15 - \frac{3}{x^4} - \frac{45}{x^2} \right)$

☐ $= \boxed{5 + \frac{14}{x^2} + \frac{9}{x^4}}$

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☐ (b) $f(t) = \frac{\sin t}{1 + \tan t}$

☐ $f'(t) = \frac{\left(\frac{d}{dt} \sin t \right) \cdot (1 + \tan t) - \sin t \left(\frac{d}{dt} (1 + \tan t) \right)}{(1 + \tan t)^2}$

☐ $= \frac{(\cos t + \sin t) - \sin t \cdot \sec^2 t}{1 + 2 \tan t + \tan^2 t}$

☐ $= \boxed{\frac{\cos t + \sin t - \sin t \cdot \sec^2 t}{\sec^2 t + 2 \tan t}}$

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$$(c) f(x) = (4x+3)^4$$

$$f'(x) = \frac{d}{dx} (4x+3)^4 \cdot (x+1)^3 - \left(\frac{d}{dx} (x+1)^3 \right) \cdot (4x+3)^4$$

$$= 16(4x+3)^3 (x+1)^2 - 3(x+1)^2 (4x+3)^4$$

$$= 16(4x^2+7x+3)^3 - 3((x+1)(4x+3)^2)^2$$

$$= \frac{16(4x^2+7x+3)^3 - 3(16x^3+40x^2+33x+9)^2}{(x+1)^6}$$

atau (korelasi $(x+1)$)

$$= \frac{(4x+3)^3 (4x+7)}{(x+1)^4}$$

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$$(d) f(x) = \sqrt{\sin(1+x^2)}$$

$$f'(x) = \frac{1}{2\sqrt{\sin(1+x^2)}} \cdot \left(\frac{d}{dx} \sin(1+x^2) \right)$$

$$= \frac{1}{2\sqrt{\sin(1+x^2)}} \cos(1+x^2) \cdot 2x$$

$$= \frac{x \cos(1+x^2)}{\sqrt{\sin(1+x^2)}}$$

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$$2. f(x) = |x^2 - 4|$$

(a) Untuk sebuah fungsi berbentuk $|g(x)|$, turunannya tidak ada pada $\forall x \geq g(x) = 0$ karena lancip.

Dengan demikian,

$$g(x) = 0 \Rightarrow x^2 - 4 = 0$$

$$\Rightarrow (x+2)(x-2) = 0$$

$$\Rightarrow x = \{-2, 2\}$$

Oleh karena itu, $|x^2 - 4|$ mempunyai turunan pada

$$X = \{x \mid x \in \mathbb{R}, x \neq 2, x \neq -2\}$$

$$\text{atau } \mathbb{R} \setminus \{-2, 2\}$$

Rumus turunannya dapat dipecah dengan

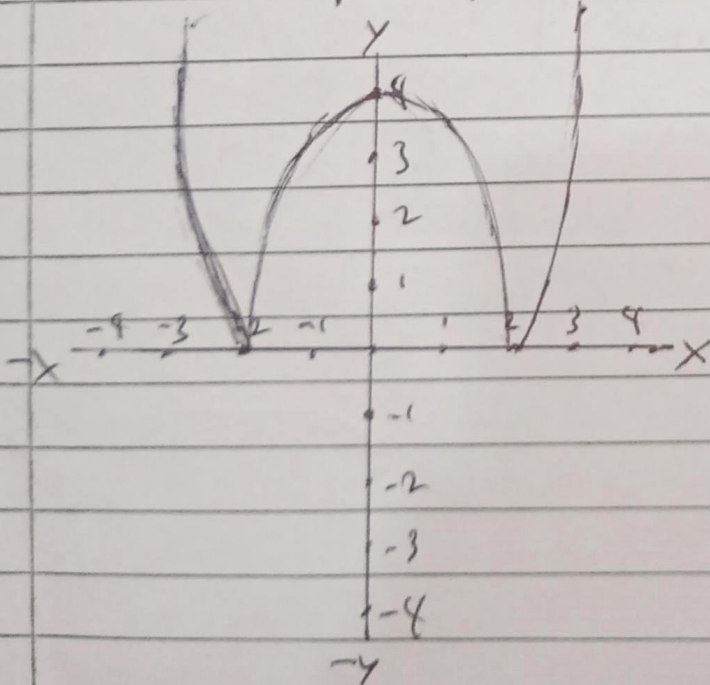
$$|x^2 - 4| = \begin{cases} x^2 - 4, & |x| \geq 2 \\ 4 - x^2, & |x| < 2 \end{cases}$$

Maka, turunannya

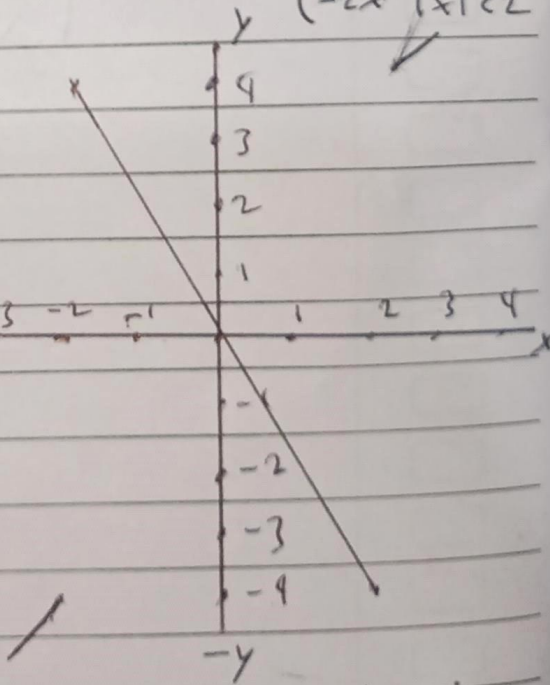
$$\frac{d}{dx} |x^2 - 4| = \begin{cases} 2x, & x \in (-\infty, -2] \cup [2, \infty) \\ -2x, & x \in (-2, 2) \end{cases}$$

(b) Menggambar grafik $f(x)$ dan $f'(x)$

$$f(x) = |x^2 - 4|$$



$$f'(x) = \begin{cases} 2x, & |x| \geq 2 \\ -2x, & |x| < 2 \end{cases}$$



$$3. f(1) = 4, g(1) = 1$$

$$f'(1) = 2, g'(1) = 6$$

$$h(x) = \frac{f(x)}{g(x) - f(x)}$$

$$g(x) - f(x)$$

$$h'(1)?$$

$$h'(x) = \frac{f'(x)(g(x) - f(x)) - f(x)(g'(x) - f'(x))}{(g(x) - f(x))^2}$$

$$\Rightarrow h'(1) = \frac{f'(1)(g(1) - f(1)) - f(1)(g'(1) - f'(1))}{(g(1) - f(1))^2}$$

$$\Rightarrow = \frac{2(1 - 4) - 4(6 - 2)}{(1 - 4)^2}$$

$$= \frac{2(-3) - 4(4)}{3^2}$$

$$= \boxed{\frac{-22}{9}}$$

$$4. y(x) = (x^2 + 1)(\sin(2x) - 1); (0, -1)$$

$$y'(x) = (2x)(\sin(2x) - 1) + (x^2 + 1)(2\cos(2x))$$

$$y'(0) = 2 \Rightarrow m = 2 \text{ untuk } (0, -1)$$

$\Rightarrow$ garis singgungnya adalah

$$\boxed{y = 2x - 1}$$

$$5. f(x) = \begin{cases} x^2 & x \leq 2 \\ mx + c & x > 2 \end{cases}$$

Karena x^2 dan $mx + c$ kontinu, agar f memiliki turunan di seluruh bilangan real, turunan dari kiri dan dari kanan dari $f(x)$ harus sama, maka

$$f'(x) = \begin{cases} 2x & x \leq 2 \\ m & x > 2 \end{cases}$$

$$\text{Maka, agar } \lim_{x \rightarrow 2^-} f'(x) = \lim_{x \rightarrow 2^+} f'(x), m = 2(2)$$

$$= 4$$

Dengan demikian, bentuk fungsinya

$$f(x) = \begin{cases} x^2 & x \leq 2 \\ 4x + c & x > 2 \end{cases}$$

Karena $f(x)$ punya turunan di $x=2$,

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$$

$$\Rightarrow [x=2] \quad x^2 = 4x + c$$

$$\Rightarrow \quad 4 = 8 + c$$

$$\Rightarrow \quad c = -4$$

Oleh karena itu

$$\underline{m = 4 \text{ dan } c = -4}$$

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$$6. f(1) = \frac{\pi}{4}, f'(1) = \frac{1}{2}$$

$$F(x) = \tan^2(f(x))$$

$$f'(x) = f'(x) \sec^2(f(x)) \cdot 2 \tan(f(x))$$

$$\Rightarrow f'(1) = f'(1) \sec^2(f(1)) \cdot 2 \cdot \tan(f(1))$$

$$= \frac{1}{2} \cdot \sec^2\left(\frac{\pi}{4}\right) \cdot 2 \cdot \tan\left(\frac{\pi}{4}\right)$$

$$= \frac{1}{2} \cdot 2 \cdot 2 \cdot 1$$

$$= \boxed{2}$$

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